Raj Reddy

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Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies **Dallas, TX**
Associate Software Engineer *June 2025 – Present*

- Built containerized microservices for high-frequency data ingestion, processing, and storage using Podman and serverless architecture on defense sensor streams.
- Developed signal processing algorithms in Python and C++ on embedded systems, meeting strict latency and resource constraints in collaboration with hardware teams.
- Automated CI/CD pipelines with static code analysis and integration testing, reducing deployment friction across engineering teams.

Doxy.me **Charleston, SC**
Software Engineer *August 2024 – May 2025*

- Improved data throughput and model accuracy for a customer-facing clinical platform by building distributed Python and TypeScript pipelines on AWS SageMaker and Fargate, designing relational schemas in PostgreSQL that normalized high-volume session data into a queryable, auditable foundation.
- Shipped React and TypeScript frontends and Node.js backend APIs end to end on a small startup team, writing clean, well-documented code with testable abstractions that let new engineers ramp up quickly without requiring tribal knowledge.
- Reduced deployment overhead and environment drift across production by owning YAML-based CI/CD configuration and cloud infrastructure automation on AWS, enabling the team to ship reliably and frequently in a fast-paced environment.

University of Utah **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.